

Browser Extension Demonstration

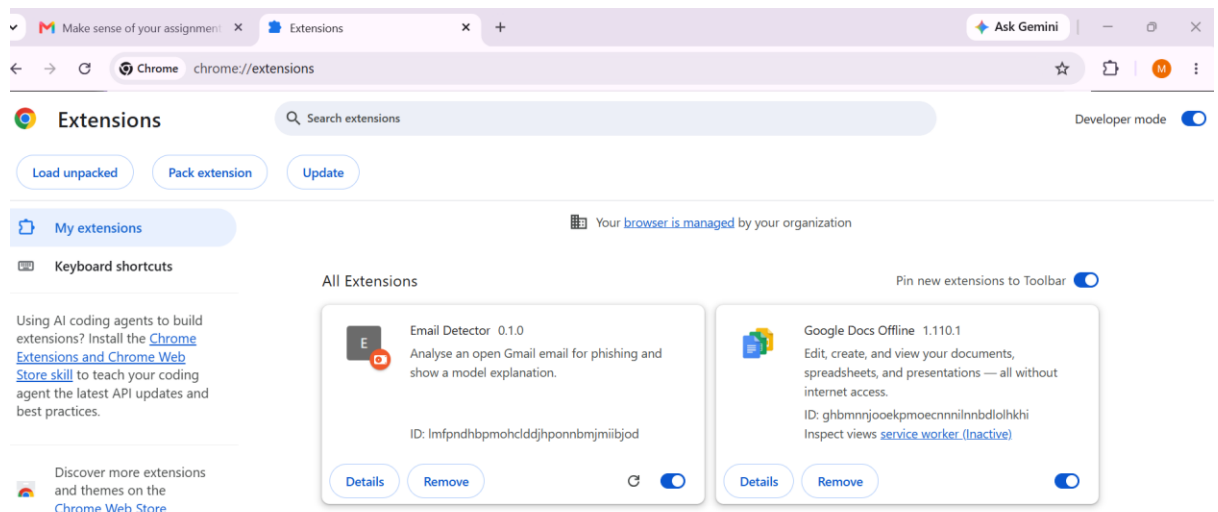
Steps for chrome extension deployment:

1. Open Google Chrome.
2. Enter `chrome://extensions` in the address bar and press **Enter**.



Google

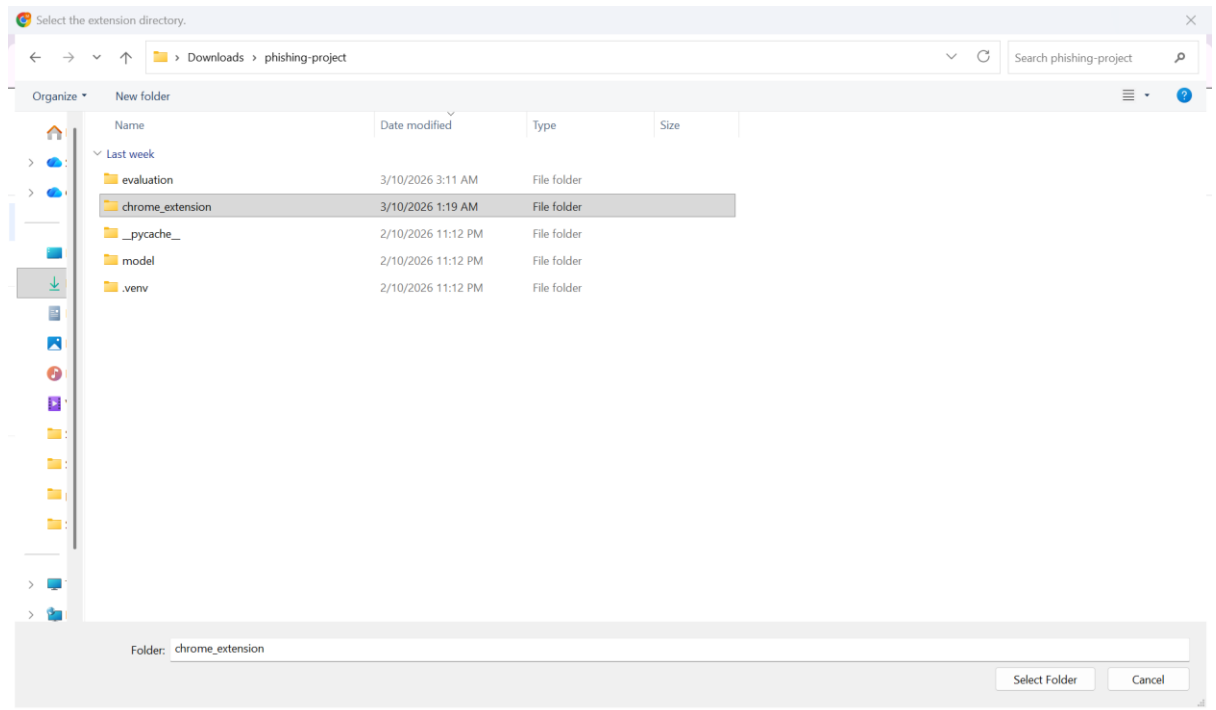
3. Turn on **Developer mode** in the top-right corner.
4. Click **Load unpacked**.



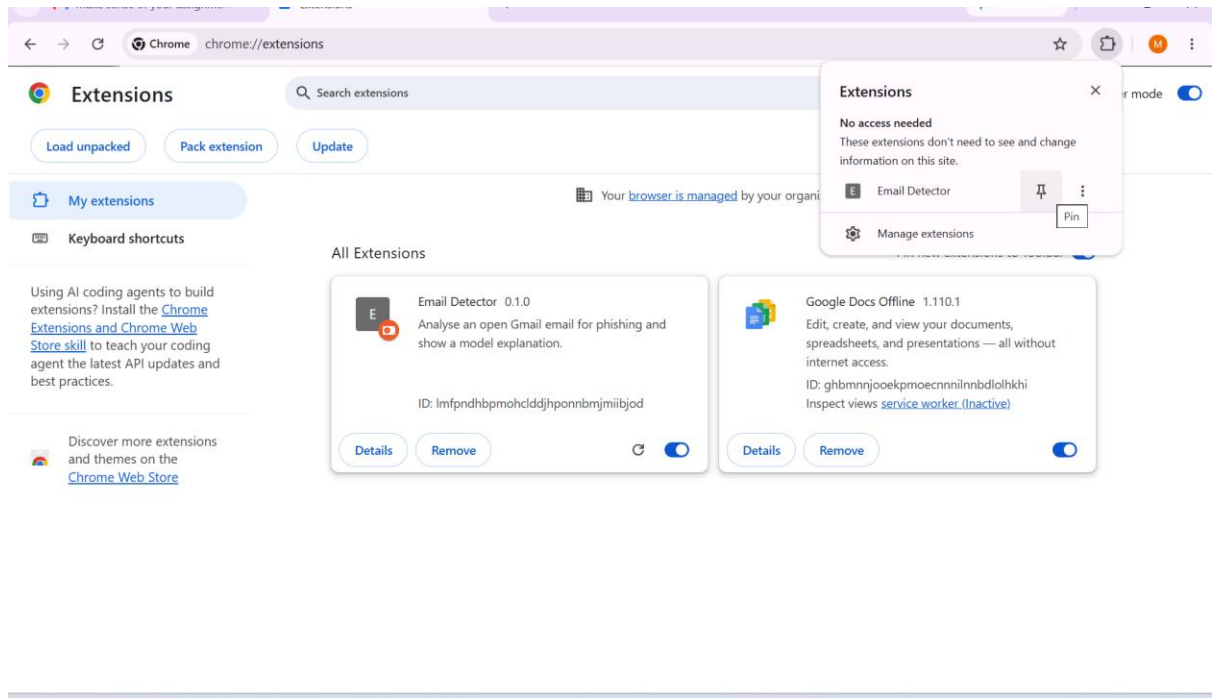
5. Select your extension folder, which is called phishing project and then select the chrome extension.

Phishing_project/ chrome-extension

6. Click **Select Folder**.



7. Confirm that **Email Detection** appears on the Extensions page without errors.
8. Click the **Extensions** puzzle icon beside the Chrome address bar.
9. Find **Email Detection** and click the **pin** icon.



10. Start the Flask backend in VS Code:

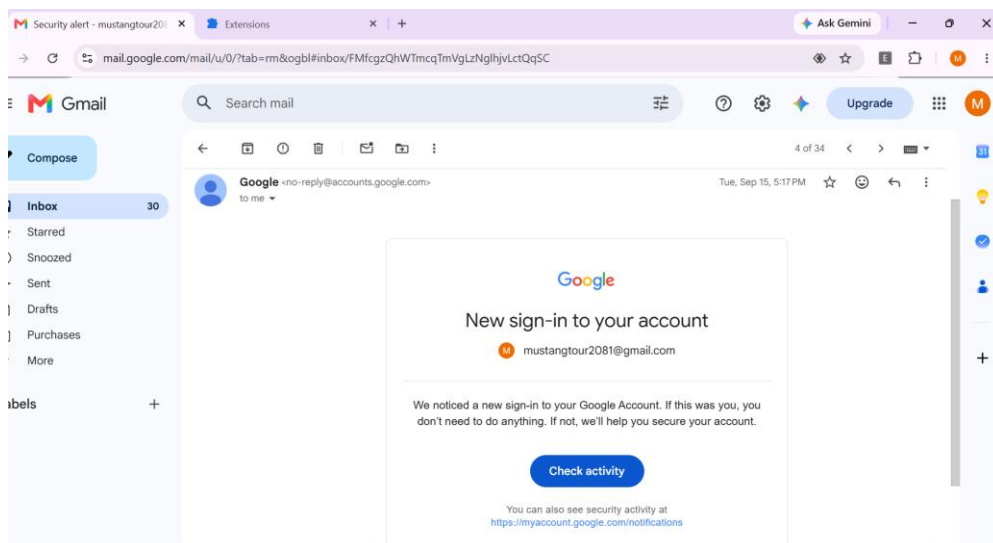
`python app.py`

11. Confirm the terminal displays:

`Running on http://127.0.0.1:5000`

12. Open Gmail in Chrome and sign in.

13. Open the email that you want to analyse.



14. Click the **Email Detection** extension icon.

 **Email Detection**

Phishing email detection

Analyse an email

Read the currently open Gmail message or paste a test email below.

Read email

Clear

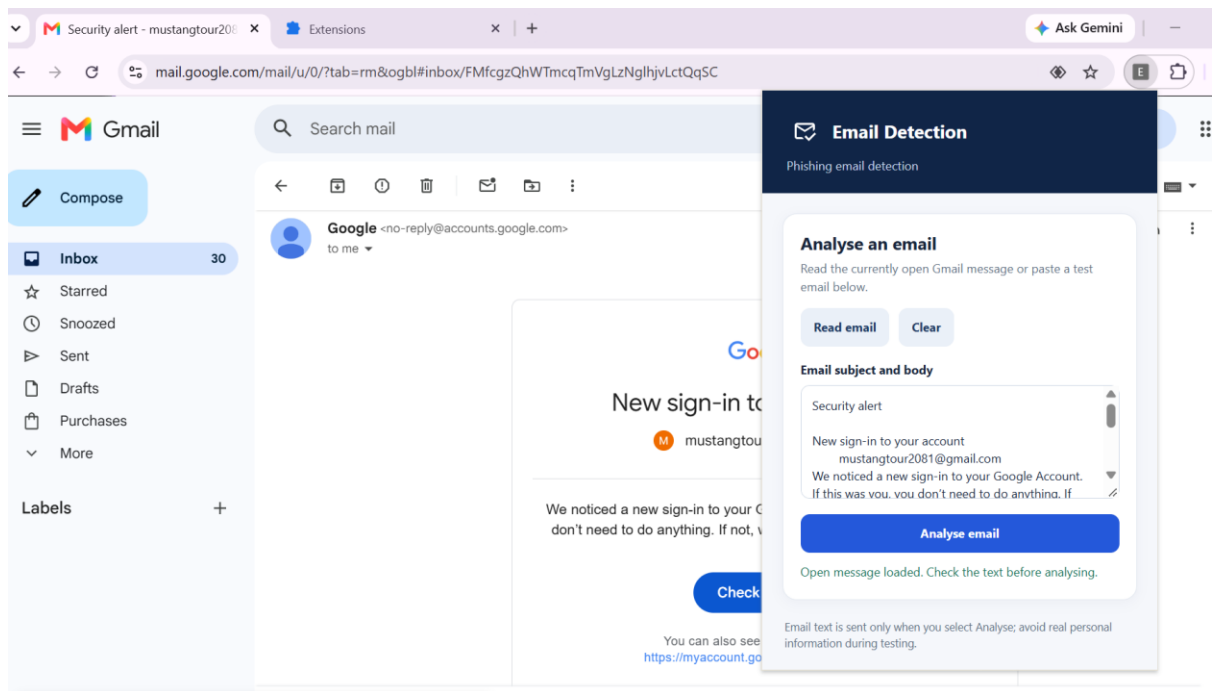
Email subject and body

Open an email in gmail and select Read, or paste a de-identified sample here.

Analyse email

Email text is sent only when you select Analyse; avoid real personal information during testing.

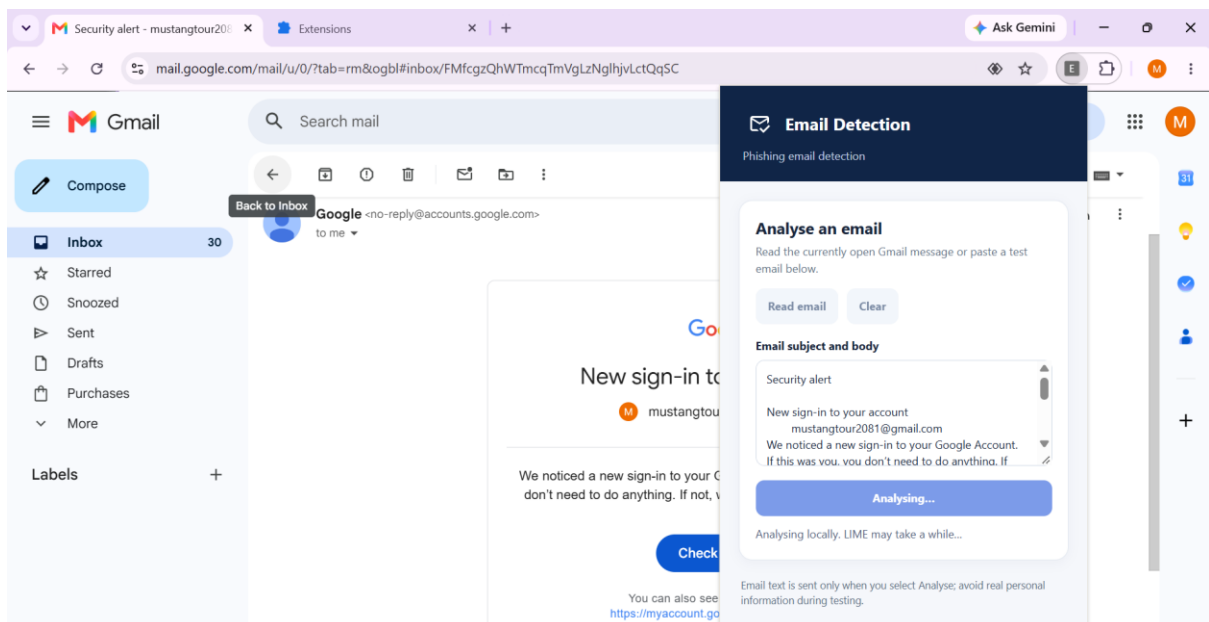
15. Click **Read email** to extract the open Gmail message.



16. Check that the email content appears in the extension text box.

17. Click **Analyse email**.

18. Wait for the extension to connect to the Flask API and process the email.



19. Review the displayed:

- Phishing or legitimate verdict
- Confidence score

○ LIME influential words and weights

Email Detection
Phishing email detection

Analyse an email
Read the currently open Gmail message or paste a test email below.

[Read email](#) [Clear](#)

Email subject and body

Security alert

New sign-in to your account
mustangtour2081@gmail.com
We noticed a new sign-in to your Google Account.
If this was you, you don't need to do anything. If not, you should check your account.

[Analyse email](#)

Analysis complete.

Analysis result

Predicted legitimate

Analysis complete.

Analysis result

Predicted legitimate

99.6%
Model score for the displayed verdict—not a guarantee.

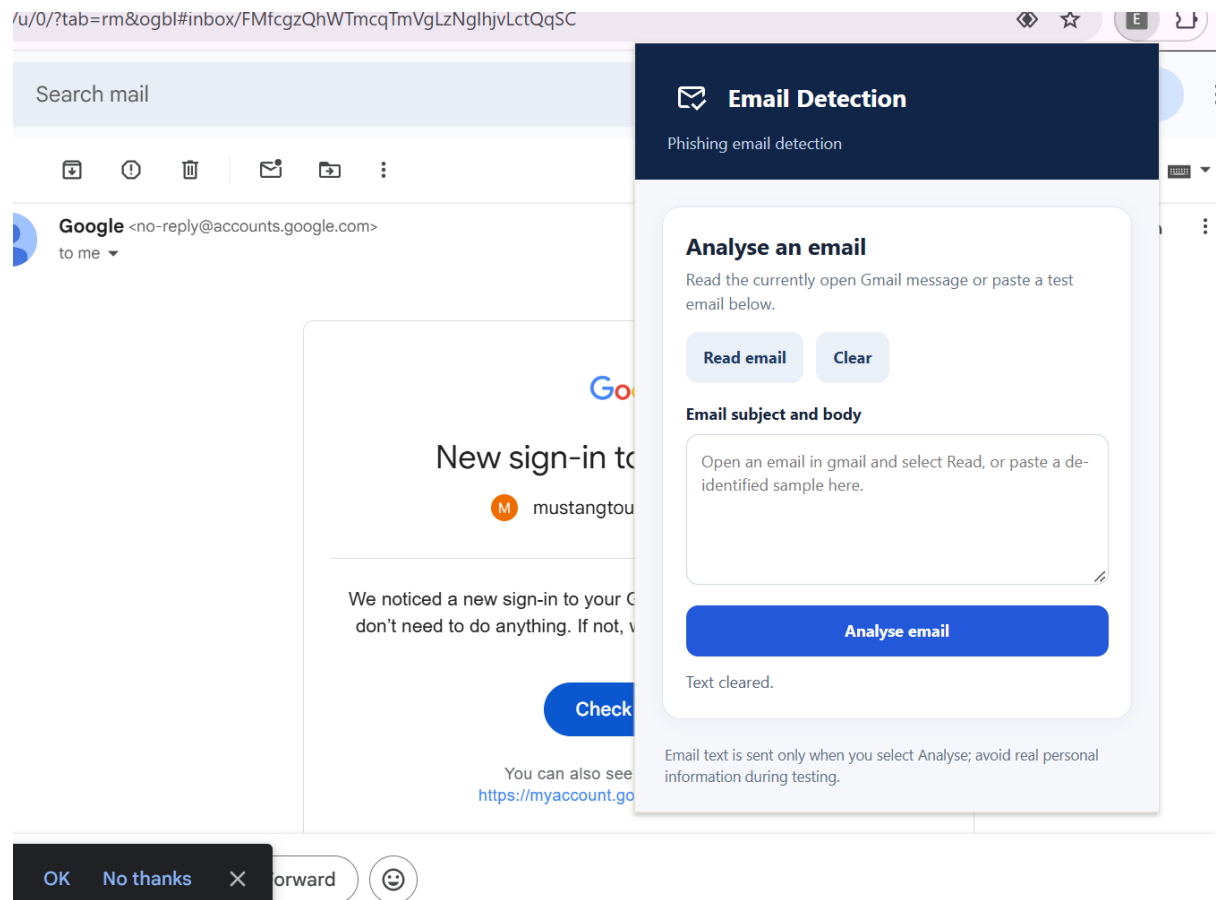
Why this result?
LIME explains the legitimate verdict. Positive weights support it; negative weights oppose it. Weights are not percentages.

Google	+0.316
You	+0.186
alert	+0.170
anything	+0.140
this	+0.140
Amphitheatre	+0.123
Check	+0.119
we	+0.003

Email text is sent only when you select Analyse; avoid real personal

ions for Gmail. [OK](#) [No thanks](#) [Forward](#)

20. To test another message, click **Clear**, open another Gmail email and repeat the process.



21. If you change manifest.json, popup.html or popup.js, return to chrome://extensions and click the extension's **Reload** button before testing again.

Testing Evidence of Legitimate and Phishing email

1. Testing legitimate emails and phishing emails

The figure consists of two screenshots of an email interface, likely Gmail, showing the analysis of a legitimate email. The email content is as follows:

Dear Sir/Madam,

Thank you for your generous contribution to the **Government of Nepal**.

Please find attached the electronically generated payment receipt.

Thank you once again for your contribution. .

Regards,
Nepal Clearing House Ltd.
www.nchl.com.np

Need Help?
For assistance, please contact our support team:
Kamladi, Kathmandu, Nepal
Phone: [+977-01-5970065](tel:+977-01-5970065)
Email: support@nchl.com.np

Email Detection
Phishing email detection

Analyse an email
Read the currently open Gmail message or paste a test email below.

Read email **Clear**

Email subject and body

Relief Fund.

Please find attached the electronically generated payment receipt for your contribution.

Thank you once again for your contribution. .

Analyse email

Analysis complete.

Analysis result

Predicted legitimate

Analysis result

Predicted legitimate

99.9%
Model score for the displayed verdict—not a guarantee.

Why this result?
LIME explains the legitimate verdict. Positive weights support it; negative weights oppose it. Weights are not percentages.

Please	+0.259
Fwd	+0.239
Clearing	+0.219
www	-0.175
message	+0.164
Wed	+0.156
not	+0.126
Phone	-0.096

Email text is sent only when you select Analyse; avoid real personal information during testing.

Figure 1: Legitimate result

Email Detection

Phishing email detection

Analyse an email

Read the currently open Gmail message or paste a test email below.

Read email

Clear

Email subject and body

Your online statement for the account ending in 1234 is now ready to view online.

Next steps

You can view your new statement in the CommBank app or NetBank.

Take the time to read your statement as it may contain important information about your account.

You can search in the CommBank app or in NetBank to find your statement.

Keep your contact details and statement

Analyse email

Analysis complete.

Analyse email

Analysis complete.

Analysis result

Predicted legitimate

100.0%

Model score for the displayed verdict—not a guarantee.

Why this result?

LIME explains the legitimate verdict. Positive weights support it; negative weights oppose it. Weights are not percentages.

search

+0.262

app

+0.237

NetBank

+0.197

CommBank

+0.172

updates

+0.130

security

+0.127

information

-0.038

Contact

+0.030

Figure 2: Legitimate emails

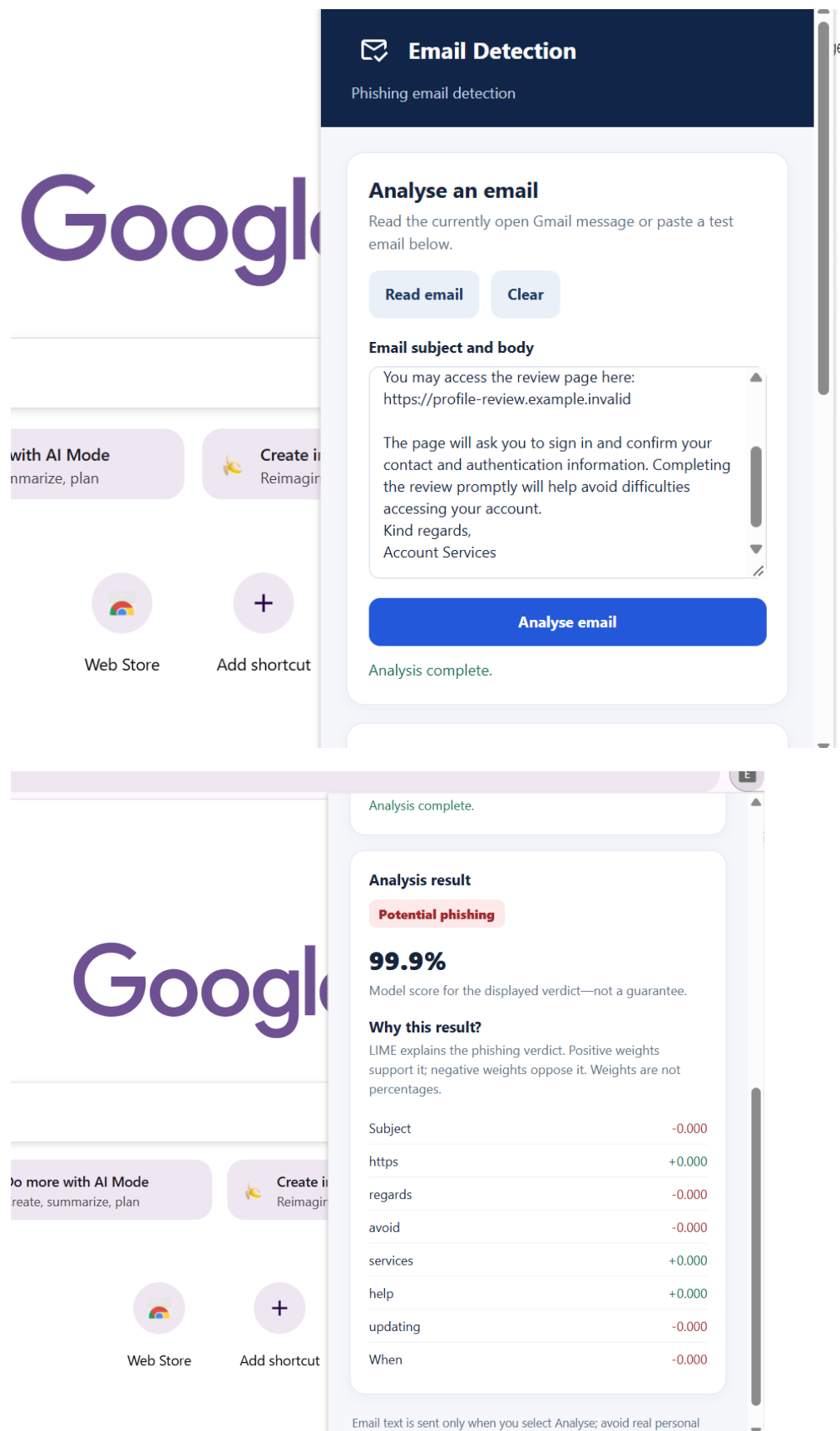


Figure 3: Phishing emails

Phishing email detection

Analyse an email

Read the currently open Gmail message or paste a test email below.

Read email

Clear

Email subject and body

Subject: URG3NT ACCOUNT NOTICE
Y0ur acc0unt requires immediate attenti0n. C!ck
the address below and c0nfirm your sign-in
information:
hxtps://secure-check[.]example[.]invalid

Failure to respond within 60 minutes may cause
account deactivation.
Support Team

Analyse email

Analysis complete.

Analysis result

Potential phishing

Analysis result

Potential phishing

100.0%

Model score for the displayed verdict—not a guarantee.

Why this result?

LIME explains the phishing verdict. Positive weights support it; negative weights oppose it. Weights are not percentages.

Failure	-0.000
respond	-0.000
acc0unt	-0.000
c0nfirm	-0.000
to	+0.000
NOTICE	+0.000
the	-0.000
may	-0.000

Email text is sent only when you select Analyse; avoid real personal information during testing.

Figure 4 : Phishing email

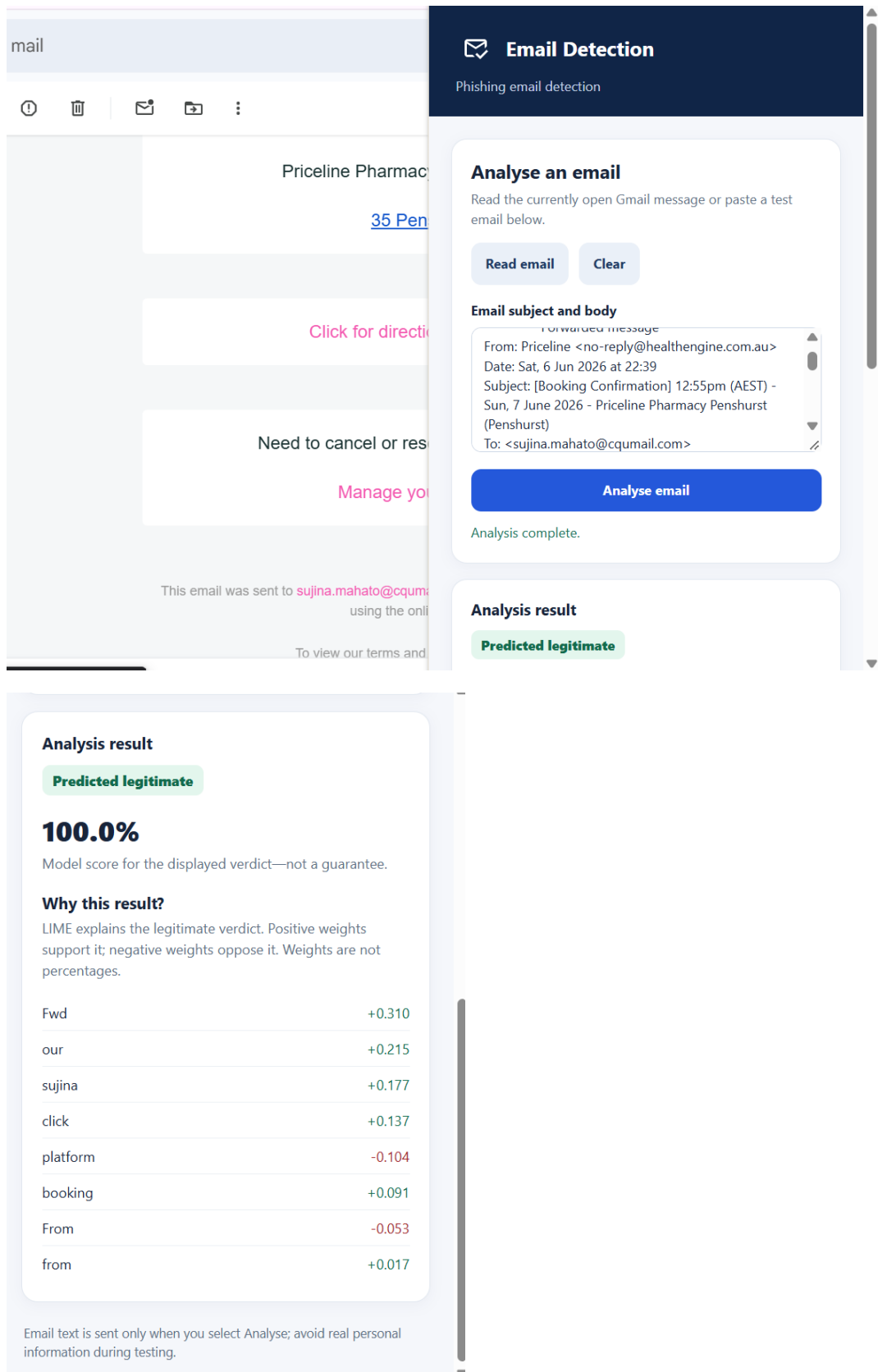


Figure 5 legitimate email

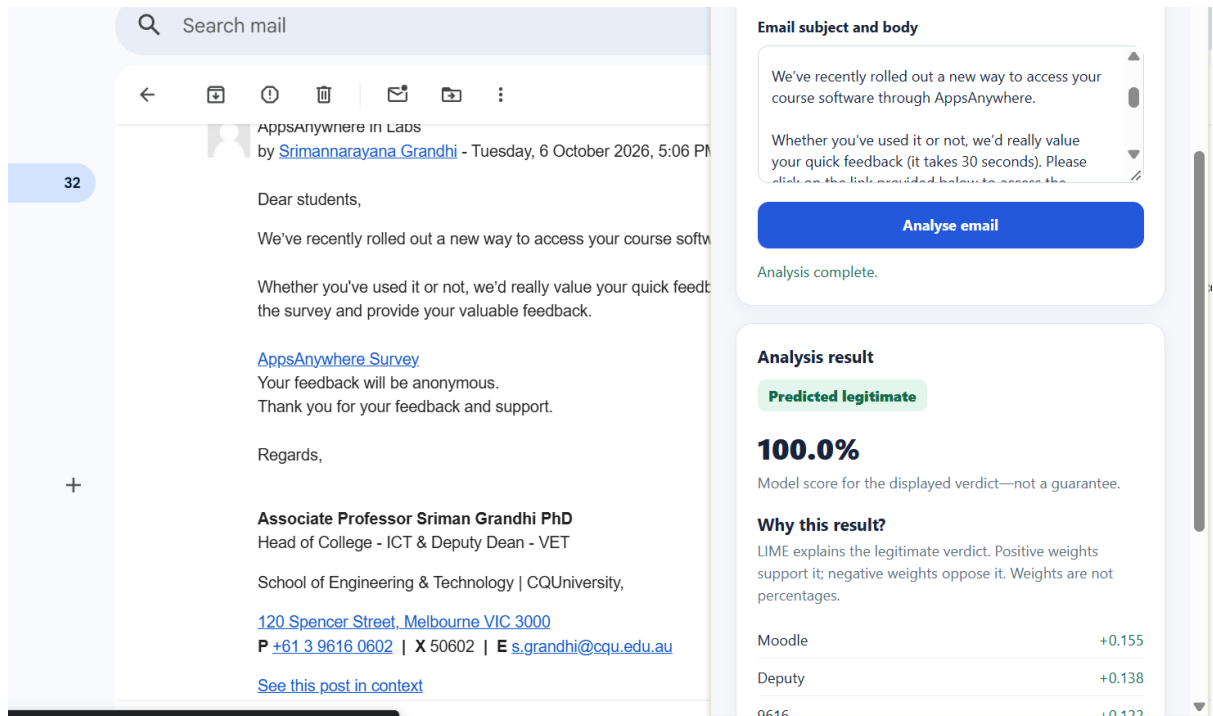


Figure 6: Legitimate email

Phishing email detection

Analyse an email

Read the currently open Gmail message or paste a test email below.

Read email

Clear

Email subject and body

Subject: Security notice
We noticed an unfamiliar sign-in attempt. To keep your account active, please v3rify your identity and upd@te your passw0rd.
Secure portal:
hxxps://account-securitv[.l]example[.]invalid

Analyse email

Analysis complete.

Analysis result

Potential phishing

99.2%

Model score for the displayed verdict—not a guarantee.

Figure 7: Phishing email

Email Detection

Phishing email detection

Analyse an email

Read the currently open Gmail message or paste a test email below.

Read email

Clear

Email subject and body

Good morning,
Just a friendly reminder that applications for the PASS Leader position close at midnight Sunday 21 September! If you are interested in joining the team and helping your peers succeed, don't miss out on this opportunity.
Make sure to submit your application before the deadline.
If you have any questions, please do not hesitate to reach out.
Note: If you have already submitted an application,

Analyse email

Analysis complete.

Analysis result

Predicted legitimate

100.0%

Model score for the displayed verdict—not a guarantee.

Why this result?

LIME explains the legitimate verdict. Positive weights support it; negative weights oppose it. Weights are not percentages.

CQUniversity	+0.238
the	+0.135
and	+0.118
disregard	-0.111
submit	+0.093
a	+0.064
do	-0.037
application	+0.004

Email text is sent only when you select Analyse; avoid real personal information during testing.

Figure 8: University legitimate mail

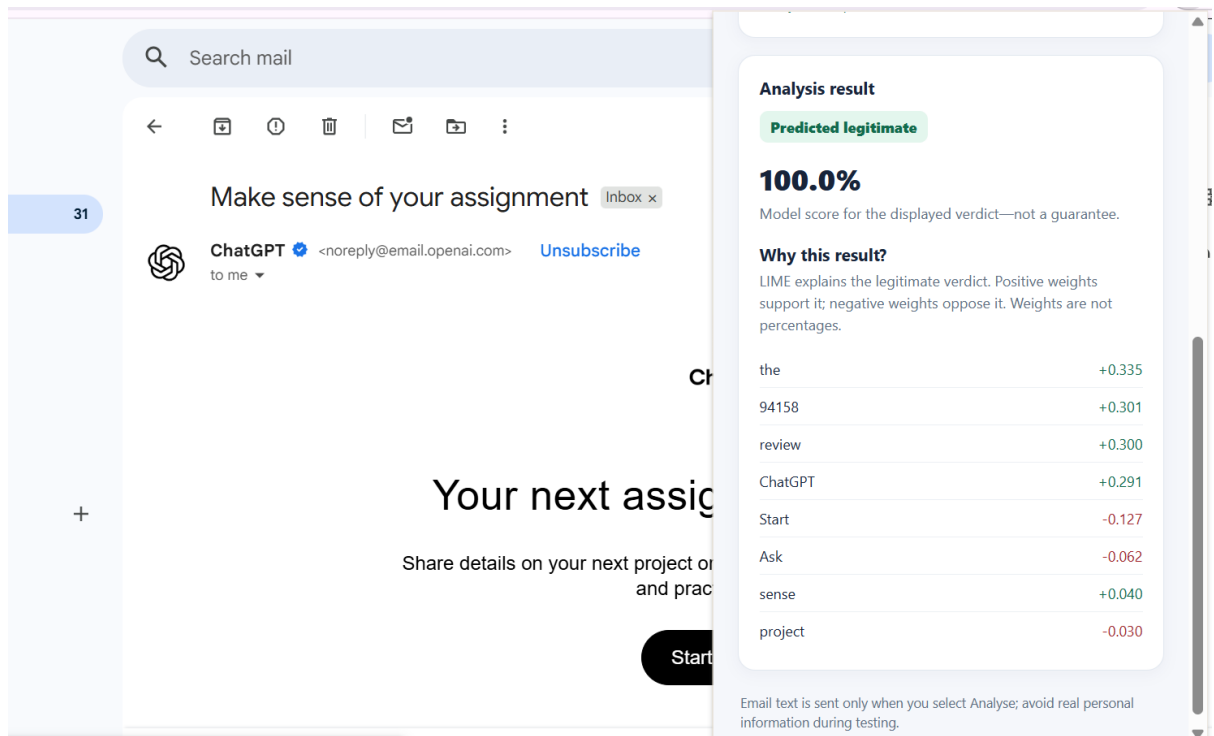


Figure 9: Legitimate



Email Detection

Phishing email detection

Analyse an email

Read the currently open Gmail message or paste a test email below.

Read email

Clear

Email subject and body

Subject: Tax refund available
You are eligible to receive a tax refund of \$786.40. We require your banking information before the refund can be deposited.
Submit your refund request here:
<https://tax-refund.example.invalid>
Your request must be completed within 48 hours.
Refund Processing Team

Analyse email

Analysis complete.

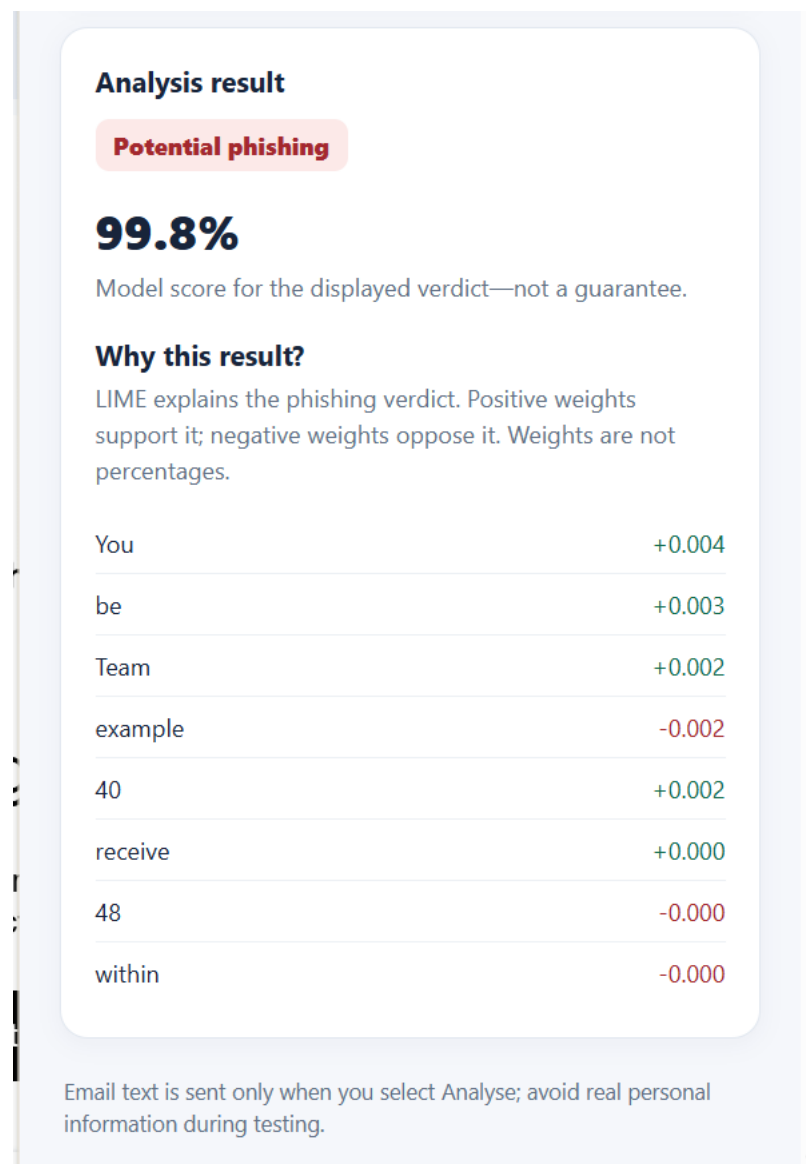


Figure 10: Phishing